

INLET RAW GAS SEPARATOR

IFC	1	22-May-23	ISSUED FOR CONSTRUCTION	LD	MMAR	IC
IFA	0	24-Jun-22	ISSUED FOR APPROVAL	MTOK	MMAR	IC
Status	Rev. n.	Date	Description	Prepared	Checked	Approved
	Rev. Index					
Client  SOUND ENERGY Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco				Project Title TENDRARA PROJECT		
Contractor  ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742				Provision of a leased train for production of LNG for Sound Energy		
Engineering  ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742						
Technology Vendor   TECHNICAL AMERICA INC 301 N Smith Avenue, 92880 Corona (CA)-USA Ph. +1 951 272 9540						
Facility Name TENDRARA			Location MOROCCO		Scale A4	Sheet of Sheets 1 of 5
Document Title INLET RAW GAS SEPARATOR				Doc. Code PRO11-ING-FR03		
				Vendor Doc. code 21044-PR-DSH-0V201		
Software: Microsoft Excel				File name		PRO11-ING-FR03

Technology vendor		DOC TITLE		INLET RAW GAS SEPARATOR		REV	
		TG DOC NUMBER		21044-PR-DSH-0V201		1	
Customer/ Engineering		ITF DOC NUMBER		PRO11-ING-FR03		PAGE	
		PROJECT		TENDRARA MICRO LNG PLANT		2	

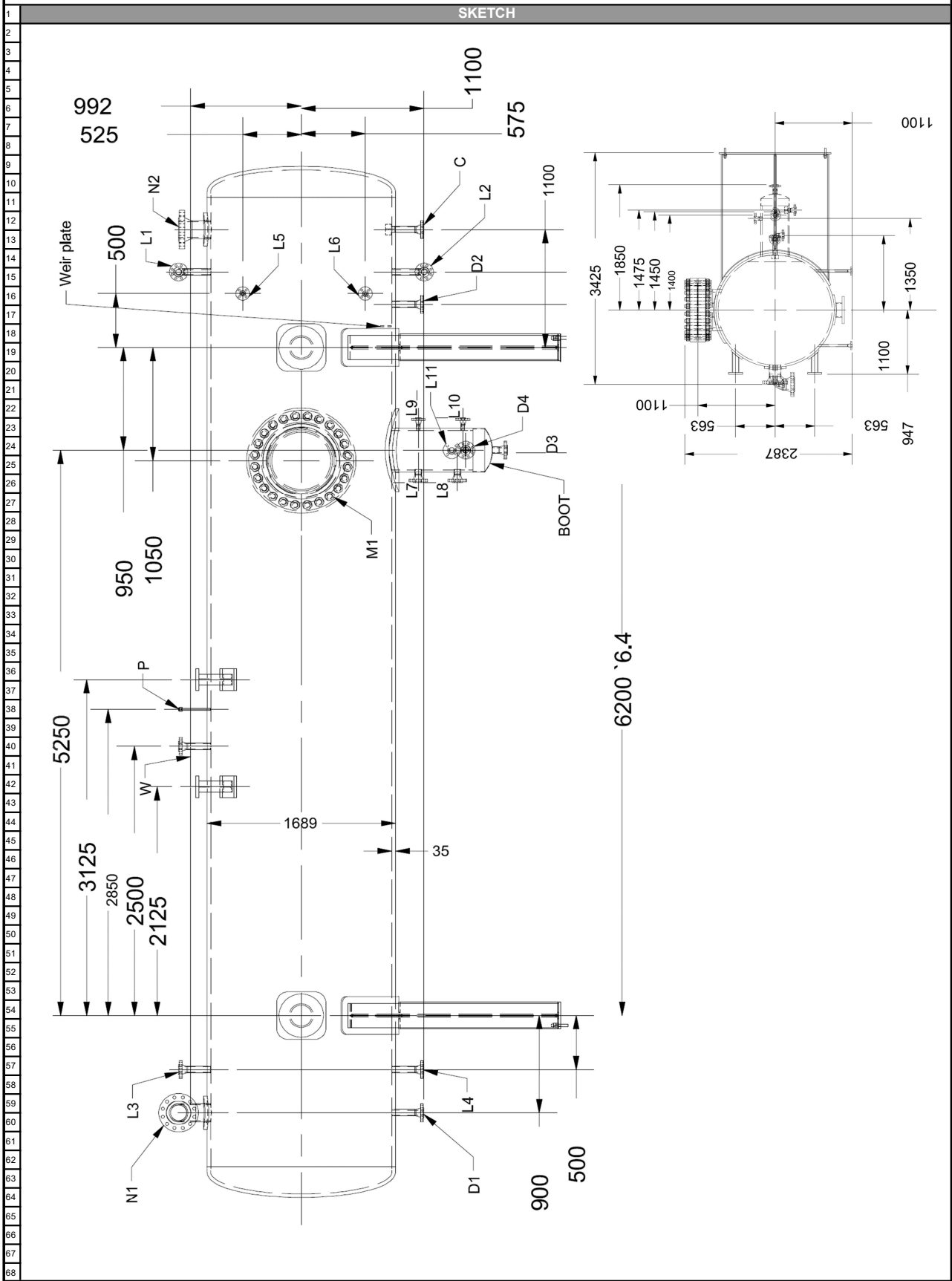
1	Plant/Process Unit Name	LNG Tendrara plant / UNIT 200		Location	Tendrara	
2	Equipment Service	Inlet raw gas separator		Quantity of Units	1	
3	Equipment TAG	V-201		P&ID	PRO13-ING-PR01	
4	Applicable to	☐ Proposal ☒ Construction		As Built	☐	
5	Model/Size and Type	3-phase horizontal		Purchaser/Agent	TG	
6				Manufacturer/Supplier	CALORIFER	
7						
8	GENERAL					
9	Head Type:	Torispherical		Volume:	19,79	m3
10	Internal Sandblast/ Paint:	No		Empty Weight:	19.500,0	kg
11	External Sandblast/ Paint/Coating:	Yes - Note 3		Hyd. Test Weight:	38.000,0	kg
12	Insulation:	No		Operating Weight:	20.000,0	kg
13	Tracing:	No				
14	Fireproof:	No				
15						
16						
17	OPERATING CONDITIONS					
18	Operating Pressure:	40,0	barg	Heavy Liquid	Design Case	Design 100% + 10% Overdesign
19	Operating Temperature norm./max.:	9,7/60	°C		Act. Volum. Flow Rate:	0,05 m3/h
20	Gas	Design Case	Design 100% + 10% Overdesign		Mass Flow Rate:	48,99 kg/h
21		S. Volum. Flow Rate:	18100+10% Sm3/d		Density:	961,1 kg/m3
22		Mass Flow Rate:	39,8 kg/h		Viscosity:	1,11 cP
23		Density:	0,0123 cP		Surface Tension:	67,9 Dyn/cm
24		Viscosity:		Foaming:	-	
25	Light Liquid:	Design Case	Design 100% + 10% Overdesign	Solids:	-	
26		Act. Volum. Flow Rate:	0,57 m3/h	CO2/H2S Content at	9,31/0	%mole
27		Mass Flow Rate:	428,9 kg/h	Flow Rates Operation Range:	-	%
28		Density:	756,0 kg/m3	Liquid (Light/Heavy) Residence Time:	>5 / >5	min
29		Viscosity:	1,05 cP			
30		Surface Tension:	19,5 Dyn/cm			
31						
32	DESIGN CONDITIONS					
33	Design Pressure:	50,0	barg @ 100 °C			
34	MDMT	-28,0	°C @ 50 barg			
35	Corrosion Allowance:	0,0	mm			
36	Design Code:	ASME Sec VIII Div.1				
37	Code Stamp Required:	No				
38						
39	BASIC DESIGN DATA					
40	Pressure Design Code	ASME VIII div. I or equivalent EN		Structural Code	As per manufacturing std	
41	Tube Bundle Code Stamped	☐ YES ☒ NO		Radiography Required	RT1	
42	Heating Coil Code Stamped	☐ YES ☒ NO		H2S Service	No	
43	Flammable/Toxic Service	☒ YES ☐ NO		Heat Treatment Required	No	
44	Lethal Service	☐ YES ☒ NO		Customer Specification	--	
45	Site Dry air Temperature	min -15 °C	max 47 °C	Design seismic action	ASCE 7-10 Site Class D	
46	Site black bulb Temperature	80 °C		Imp.Factor=1, Ss=0.175g, S1=0.088g, Risk Cat. IV		
47	Relative humidity	min 18 %	max 70 %	Design wind action	50 m/s (maximum) A	
48	Site elevation	1434 m asl		Barometric pressure	1014 mbar	1029 mbar
49	Equip elevation to plant	0,3 m				
50	INTERNALS					
51	Mist Eliminator:			Coalescer: Type:	-	
52	Type: -	Shape: -		Size	-	
53	Size	-	mm	Thickness:	- mm	
54	Thickness:	-	mm	Coil: Design Pressure	- barg @ -	°C
55	Dist between HHLL and bottom of mist eliminator	-	mm	Dia - mm	Spies Qty -	
56	Flow Direction:	-		Corrosion Allowance	- mm	
57						
58						
59	VESSEL DIMENSIONS			OTHER REQUIREMENTS		
60	Shell (ID x Ltt):	1625 x 9000	mm x mm	Instruments Clips Required:	No	
61	Boot (ID x H):	16"NPS SCH80 x 800	mm x mm	Ladder/Platform Clips Required:	Yes	
62	Thickness(Shell/head/boot):	32/32/21,44	mm	Other:	No	
63	Saddles (H):	2325	mm			
64						
65						
66						
67						
68						
69						

Technology vendor  		DOC TITLE	INLET RAW GAS SEPARATOR	REV
		TG DOC NUMBER	21044-PR-DSH-0V201	1
Customer/ Engineering 		ITF DOC NUMBER	PRO11-ING-FR03	PAGE
		PROJECT	TENDRARA MICRO LNG PLANT	3

CONSTRUCTION MATERIALS				
1				
2	Heads/Shell:	SA-516 Gr.70(N) + 304 cladding 3 mm	Internals:	
3	Nozzle (untill 16" NPS):	SA 312 TP304	Antiwave Baffle:	N/A
4	Nozzle (above 16" NPS):	SA-106 Gr B or SA 516 + 304 cladding	Inlet Diverter:	SA 403 WP404
5	Flanges (untill 16" NPS):	SA-182 F304 (till 12" NPS)	Vortex Breaker:	SA-240 TP304
6	Flanges (above 16" NPS)	SA-105 + 304 cladding	Weir Baffle:	SA-340 TP304
7	Fittings (untill 16" NPS):	SA 403 WP404	Mist Eliminator	-
8	Fittings (above 16" NPS):	SA-234 WPB+ 304 cladding	Coalescer:	N/A
9	Studs:	SA-193 Gr B7 FRP coated	Supports:	SA-516 Gr70
10	Nuts:	SA-194 Gr 2H FRP coated	Boltings:	SS304
11	Gaskets:	SPIRAL WOUND S304	Coil:	N/A
12	Saddles:	SA-516 Gr.70(N)	Others:	N/A
13	Lifting Lugs:	SA 516 Gr.70 (N) / SA 516 Gr.70		
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				
26				
27				
28				
29				
30				

NOZZLES							
Tag	Qty	Dia (in)	Type	Rating	Sch.	Service	Notes
N1	1	6	WNRF	#600	80S	Natural gas Inlet	w/inlet diverter
N2	1	6	WNRF	#600	80S	Natural gas outlet	-
C	1	2	WNRF	#600	80S	Light liquid outlet	w/ vortex br
W	1	2	WNRF	#600	80S	PSV	-
P	1	1/2	NPT	#6000	-	Pressure instrument	-
L1/L2	2	2	WNRF	#600	80S	Level Transmitter light liquid - Control	-
L3/L4	2	2	WNRF	#600	80S	Level Transmitter light liquid - Safety	-
L5/L6	2	1	WNRF	#600	80S	Level Gauge light liquid	-
L7/L8	2	2	WNRF	#600	80S	Level Transmitter heavy liquid - Control	-
L9/L10	2	1	WNRF	#600	80S	Level Gauge heavy liquid	-
L11	1	1 1/2	WNRF	#600	-	Level switch heavy liquid - Safety	-
D1	1	2	WNRF	#600	80S	Drain	-
D2	1	2	WNRF	#600	80S	Drain	-
D3	1	2	WNRF	#600	80S	Boot Drain	-
D4	1	2	WNRF	#600	80S	Boot- Heavy liquid out	w/ vortex br
M1	1	24	WNRF	#600	40mm	Manway	-
49							
50							
51							
52							
53							
54							
55							
56							
57							
58							
59							
60							
61							
62							
63							
64							
65							
66							
67							
68							

Technology vendor 	DOC TITLE	INLET RAW GAS SEPARATOR	REV
	TG DOC NUMBER	21044-PR-DSH-0V201	1
	ITF DOC NUMBER	PRO11-ING-FR03	PAGE
Customer/ Engineering 	PROJECT	TENDRARA MICRO LNG PLANT	4



Technology vendor  	DOC TITLE	INLET RAW GAS SEPARATOR	REV
	TG DOC NUMBER	21044-PR-DSH-0V201	1
Customer/ Engineering 	ITF DOC NUMBER	PRO11-ING-FR03	PAGE
	PROJECT	TENDRARA MICRO LNG PLANT	5

1	NOTES
2	
3	1 - Test pressure shall be witnessed by authorized Moroccan inspector to allow importation in Morocco.
4	2 - Deleted.
5	3 - See Painting procedure MEC07-ING-FR03 (21044-QC-PAI-00001).
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	
42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	
67	
68	